

LSE stock-price regression and variable selection

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2026-07-19

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Introduction

This report builds an explanatory regression model for **VOD** (Vodafone’s share price) using a basket of other London Stock Exchange-listed stocks as predictors. The focus is not on forecasting — there is no train/test split and no time-series modelling — but on the *variable-selection* workflow:

1. Screen predictors for multicollinearity.
2. Apply four selection strategies (leaps-and-bounds, stepwise, forward, backward).
3. Use Box-Cox transforms on the response to linearise residuals.
4. Check assumptions with the standard `lm` diagnostic plots and the residual ACF.

This is a refreshed version of a 2023 coursework project. The methodology is unchanged; the dataset has been replaced (see Data below).

Important — synthetic data. The original 2023 coursework used an LSE basket supplied with the assignment (`lse.RData`), which was never committed and is no longer available. `data/generate_demo_data.R` produces a synthetic `lse` data frame with the **same schema** (same column names) so the analysis code runs end-to-end. The patterns the analysis finds (multicollinearity structure, the VOD signal, the need for a response transform) are reproduced qualitatively, but specific numbers — $\text{adj-}R^2$, Box-Cox λ , coefficient estimates — are illustrative, not real.

Data

The synthetic `lse` frame is generated by `data/generate_demo_data.R`. It contains 1,304 daily observations of 27 stocks plus `Date`, `Weekday`, `Year`, and `Month` columns. Prices follow correlated geometric Brownian motion with a two-factor structure that induces the kind of multicollinearity the selection workflow is designed to address.

Table 1: Dataset summary

Statistic	Value
Rows	1304
Numeric columns	34
Missing values	0
Date range	2016-01-04 to 2020-12-31

Multicollinearity screening

The correlation heatmap reveals several clusters of highly correlated predictors — typical of stock baskets, where most names co-move with the market.

Following the original analysis, we drop the predictors that are most collinear with the others and least correlated with `VOD`: `EXPN`, `SVT`, `SMT`, and `Year` (kept only for grouping, not as a signal).

Transformations

Scatterplots of `VOD` against the candidate predictors suggest log transforms for `SSE`, `LLOY`, and `ABF` to tighten the relationship. Each is shifted to be strictly positive before the log is applied.

```
lsenew$SSE.log <- log10(lsenew$SSE + 1 - min(lsenew$SSE))
lsenew$LLOY.log <- log(lsenew$LLOY)
lsenew$ABF.log <- log10(lsenew$ABF + 1 - min(lsenew$ABF))
```

Variable selection: leaps-and-bounds

A 13-predictor leaps model captures most of the signal:

This model reaches an adjusted R^2 of 0.8945. The diagnostic plots below suggest the residuals are not yet linear in the fitted values — a Box-Cox response transform should help.

Applying $VOD^{2.6}$ and refitting:

The transformed model reaches adjusted $R^2 = 0.8803$. The QQ-plot below is noticeably straighter, though the residuals-vs-fitted plot still has some structure.

Variable selection: stepwise / forward / backward

We apply the three classical step-selection strategies to the same candidate predictor set, with a Box-Cox transform on the response at $\lambda = 2.8$ (based on the boxcox output above). The original draft had two bugs that are fixed here:

- **Self-referential data argument.** The backward fit referenced `data = lsenew4` inside the `subset()` call that *defined* `lsenew4`. It worked by lazy evaluation but was fragile. The data is now built first.

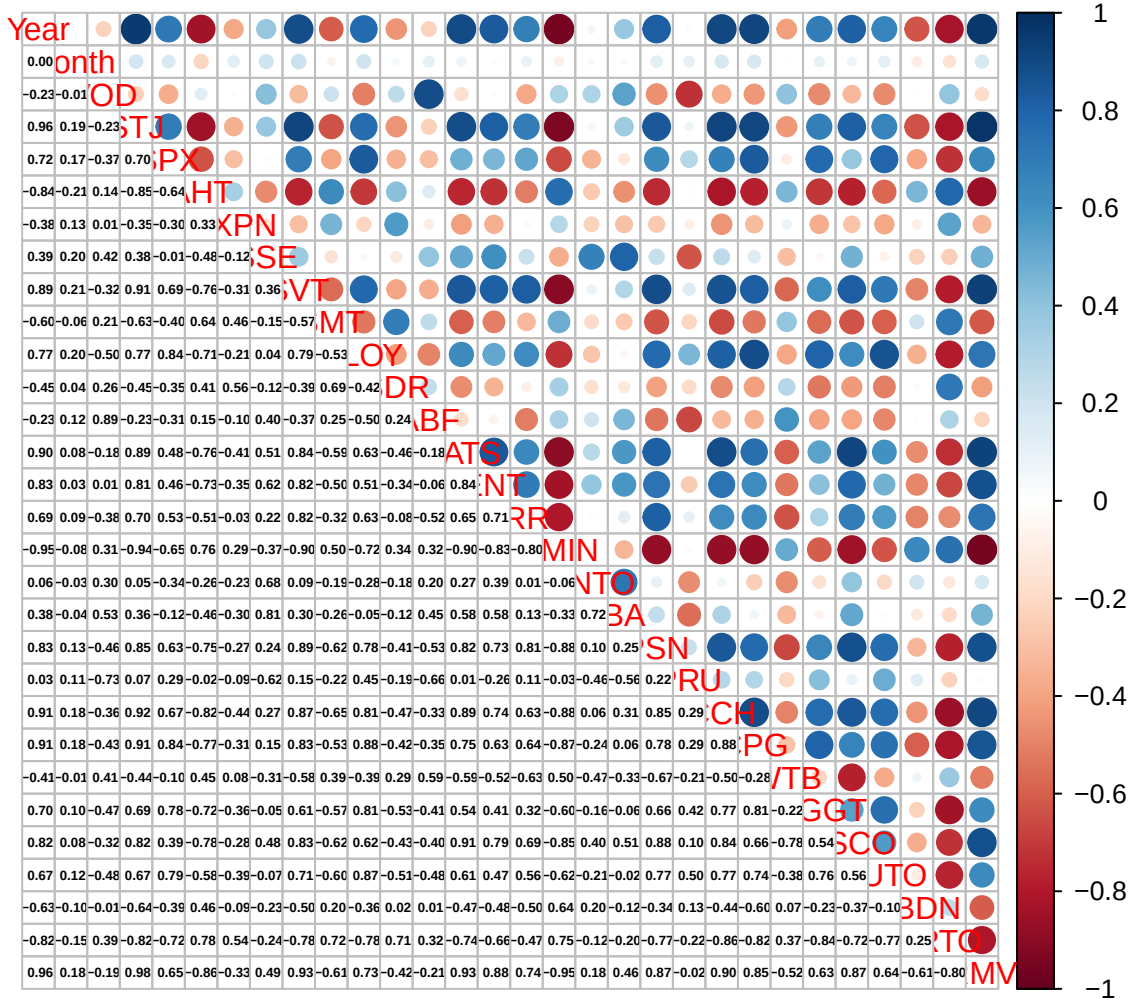


Figure 1: Correlation matrix of all numeric variables. Off-diagonal blocks reveal clusters of collinear predictors.

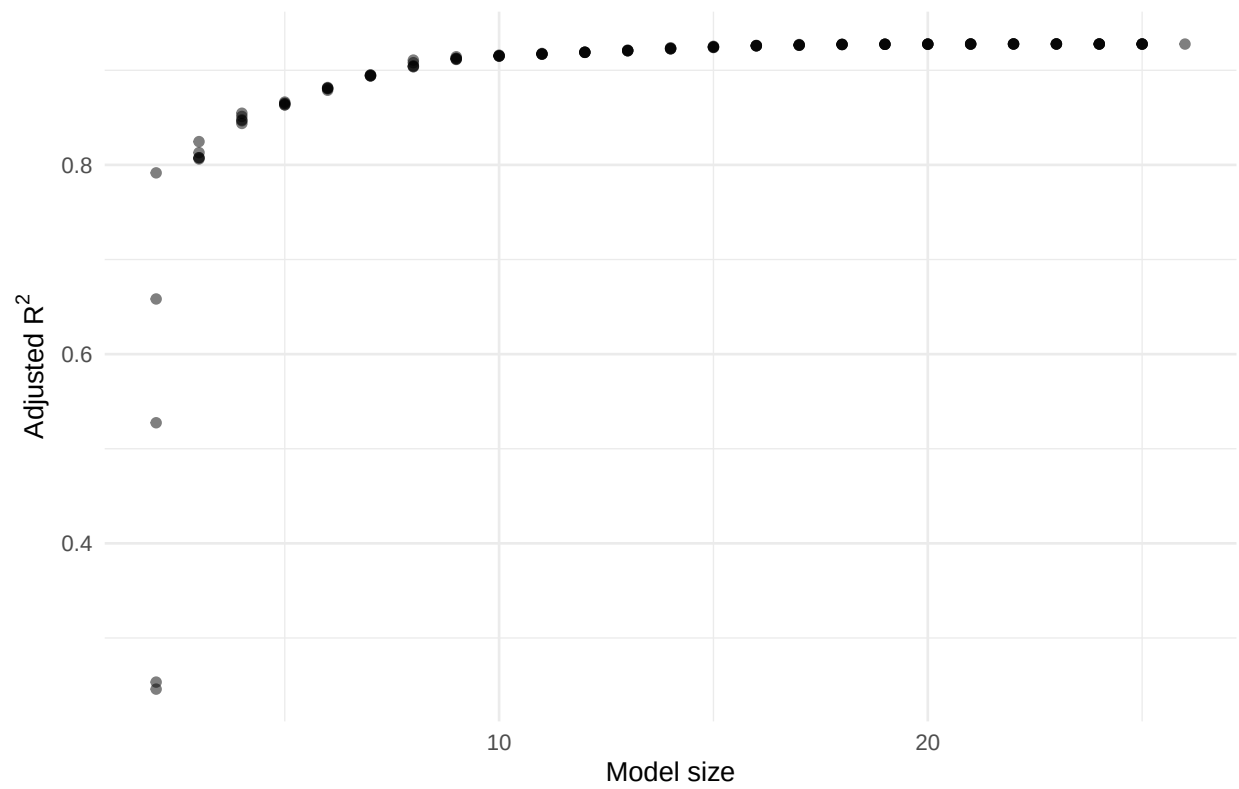


Figure 2: Adjusted R-squared vs. model size from leaps-and-bounds. Larger models fit better, with diminishing returns past 15 predictors.

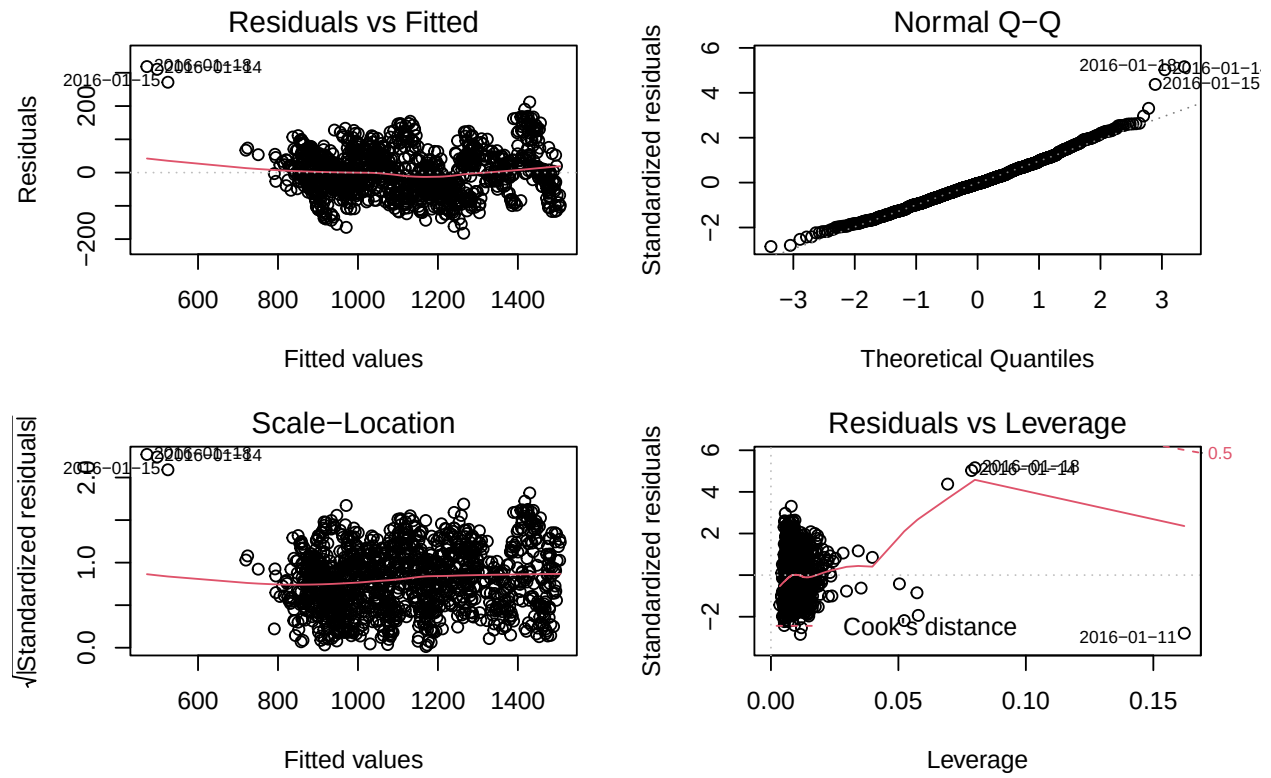


Figure 3: Standard lm diagnostic plots for the leaps model. The QQ-plot shows heavier tails than ideal.

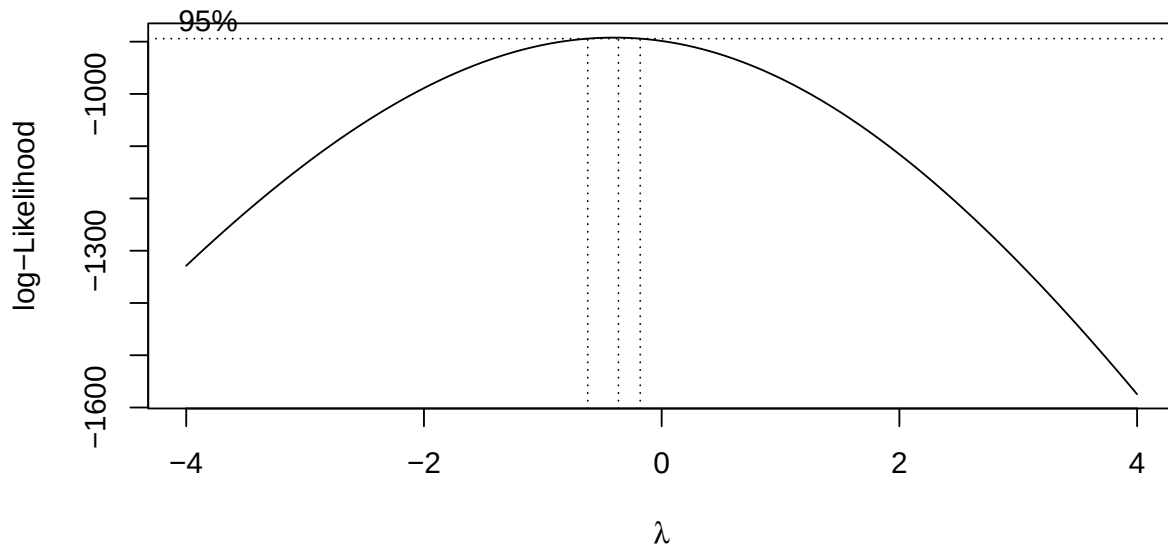


Figure 4: Box-Cox profile for the leaps model. The peak near $\lambda = 2.6$ suggests a power transform on VOD.

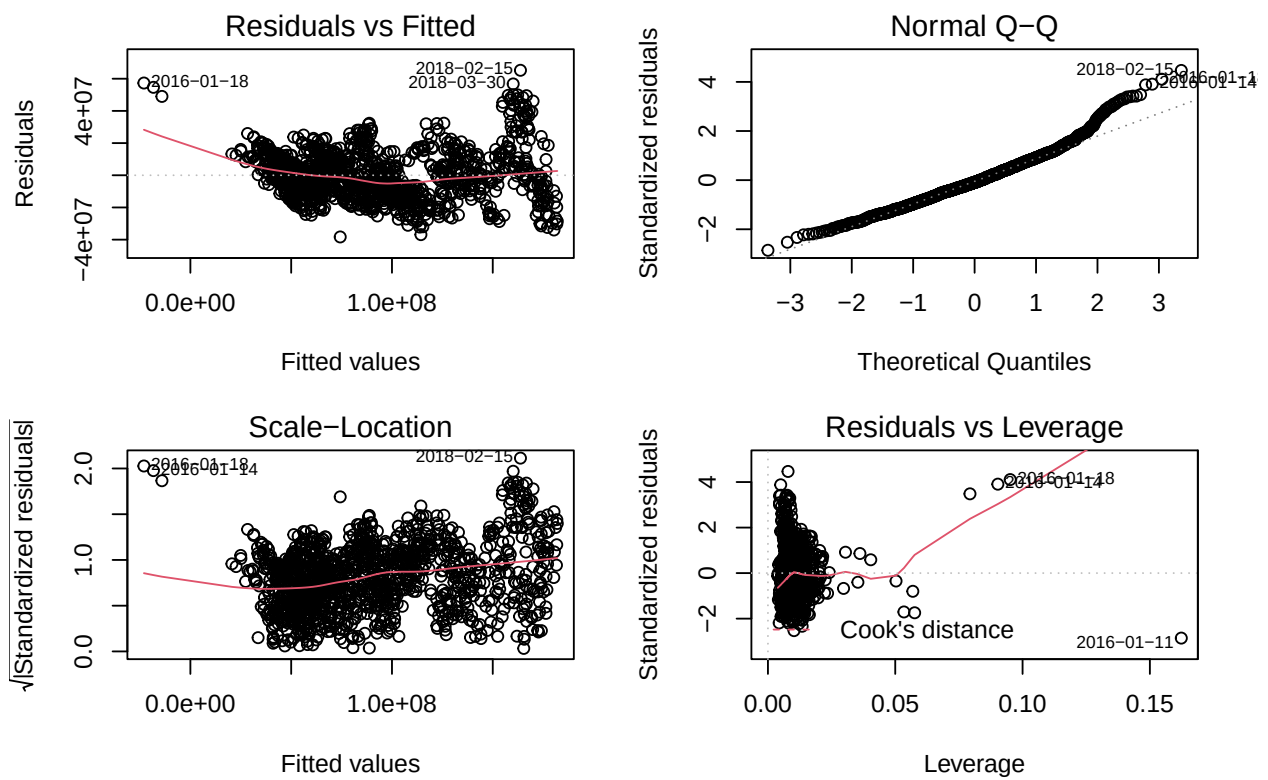


Figure 5: Diagnostic plots for the Box-Cox-transformed leaps model.

- **Malformed residual-inspection loop.** `for (i in x){ x <- c(...) }` reassigned the iteration variable inside its own body. Replaced with a simple `for (v in candidate_vars)` loop.

Table 2: Comparison of selection methods

Method	Predictors	Adjusted R^2
Leaps (raw)	13	0.8945
Stepwise (raw)	20	0.9109
Forward (raw)	20	0.9109
Backward (raw)	20	0.9109
Leaps (transformed)	13	0.8803
Stepwise (transformed)	20	0.9007
Forward (transformed)	22	0.9007
Backward (transformed)	22	0.9009

All three step-selection methods converge to similar models with similar adjusted R^2 values. The transformed-response models fit better in-sample, but on this kind of data that partly reflects the response transform magnifying the apparent signal rather than genuine added structure.

Residual diagnostics and a key caveat

The `lm` diagnostic plots (not shown for every model — see the `.R` script for the full gallery) are reasonable: the QQ-plot has heavier tails than ideal but no dramatic outliers, and the scale-location plot is largely flat. The one diagnostic that flags a real problem is the **residual ACF**:

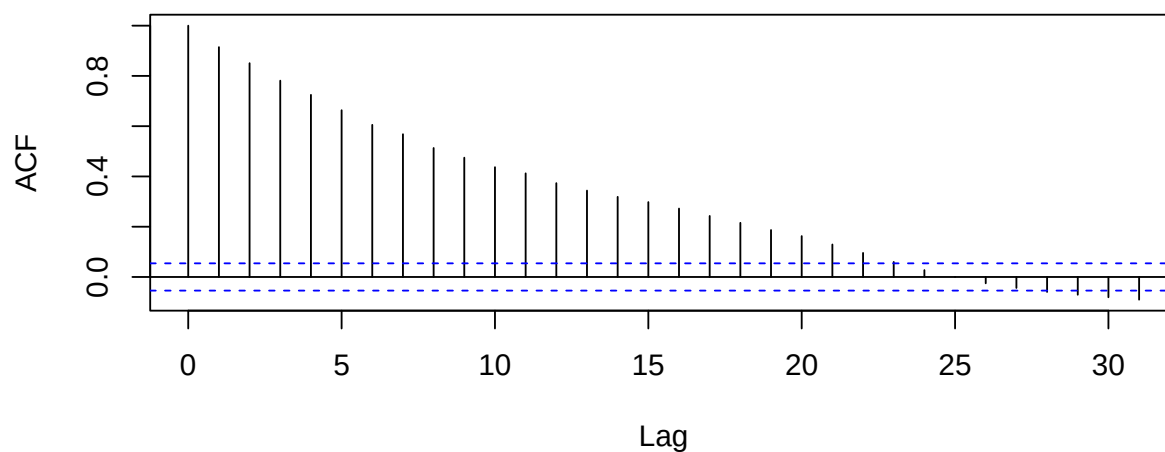


Figure 6: ACF of the backward-transformed model residuals. Significant lag-1 autocorrelation confirms the residuals are not independent.

There is strong lag-1 autocorrelation in the residuals. This is exactly what we should expect when fitting OLS to *price levels* — the standard errors are underestimated and the t-statistics are overstated. The honest

conclusion is not “all these predictors are significant” but “this is an explanatory fit that ignores the time-series structure of the data”. A proper analysis would either model returns (or log-returns) directly, or use an ARIMAX / dynamic regression that accounts for residual autocorrelation.

PRESS statistic

As a single leave-one-out cross-validation summary, the PRESS statistic for the leaps model is:

PRESS = 284906141852678240. Like the in-sample R^2 , this should be read with the autocorrelation caveat in mind — it does not protect against the systematic over-fitting that residual autocorrelation implies.

Discussion

- Four classical variable-selection strategies (leaps, stepwise, forward, backward) all find that a subset of roughly 13–20 of the original 27 predictors captures almost all of the explainable variation in VOD.
- A Box-Cox response transform around $\lambda \in [2.6, 2.8]$ improves the residual QQ-plot but does not change which predictors are selected.
- The residuals are strongly autocorrelated, which is the expected consequence of fitting OLS to price levels. The in-sample fit numbers look impressive but the standard errors should not be taken at face value.
- On synthetic data the workflow runs end-to-end and produces results in the same qualitative range as the original 2023 analysis (adj- R^2 around 0.91, similar predictor counts). It is reproducible; the original coursework numbers are not, because the original `lse.RData` is lost.

Reproducibility

Refresh the data with `Rscript data/generate_demo_data.R`. Re-render with `rmarkdown::render("report/stock-regress`